

Bugs and debugging

Programmers aren't perfect, and most programs contain errors at first. These errors are known as "bugs" and tracking them down is called "debugging".

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in Python

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Types of bugs

Three main types of bugs can turn up in programs—syntax, runtime, and logic errors. Some are quite easy to spot, while others are more difficult, but there are ways of finding and fixing them all.

The Python keyword
is "for" not "fir"

```
fir i in range(5):  
    print(i)
```

△ Easy to spot

A syntax error is a mistake in the program's words or symbols, such as misspelled keywords, missing brackets, or incorrect indents.

This will cause
an error as no
number can be
divided by 0

```
a = 0  
print(10 / a)
```

△ Harder to spot

Runtime errors appear only when the program is running. Adding numbers to strings or dividing by 0 can cause them.

Age cannot be less than 5
and greater than 8 at the
same time, so no free tickets

```
if age < 5 and age > 8:  
    print('Free ticket!')
```

△ Hardest to spot

Logic errors are mistakes in a program's thinking. Using "<" instead of ">", for example, or adding when you should be subtracting result in these errors.

Find and fix a bug

Syntax errors are easy to spot because IDLE highlights them in red when you run the program. Finding runtime and logic errors takes a bit more work.

1 Problem program

This program aims to add all the numbers from 1 up to the value stored in the variable "top_num". It then prints the total.

```
top_num = 5  
total = 0  
for n in range(top_num):  
    total = total + n  
print('Sum of numbers 1 to', top_num, 'is', total)
```

The highest number in the series of numbers being added

This command prints a sentence to let the user know the result

2 Output

The answer for the program should be (1 + 2 + 3 + 4 + 5), but it shows the answer as "10". You need to find out why.

Sum of numbers 1 to 5 is 10

The answer should be "15", not "10"

